

HIGH PERFORMANCE COMPUTING FOR SCIENCE AND ENGINEERING

Harvard University
CS 2050

Lecture 6
February 12, 2026

FINAL EXAM

- May 13, 9am

- Proctorio for CSCI E-205

Deadlines

February

Tue Feb 03 - [Quiz 1](#) due by 11:59pm

Tue Feb 10 - [Quiz 2](#) due by 11:59pm

Thu Feb 12 - [Homework 1](#) due by 11:59pm

Tue Feb 17 - Quiz 3 due by 11:59pm

Tue Feb 24 - Quiz 4 due by 11:59pm

Thu Feb 26 - Homework 2 due by 11:59pm

March

Tue Mar 03 - Quiz 5 due by 11:59pm

Tue Mar 10 - Quiz 6 due by 11:59pm

Thu Mar 12 - Homework 3 due by 11:59pm

----- Spring Recess -----

Tue Mar 24 - Quiz 7 due by 11:59pm

Tue Mar 31 - Quiz 8 due by 11:59pm

April

Thu Apr 02 - Homework 4 due by 11:59pm

Tue Apr 07 - Quiz 9 due by 11:59pm

Tue Apr 14 - Quiz 10 due by 11:59pm

Thu Apr 16 - Homework 5 due by 11:59pm

Tue Apr 21 - Quiz 11 due by 11:59pm

Tue Apr 28 - Quiz 12 due by 11:59pm

May

Wed May 13 - Final Exam at 9am

homework-1

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Code

About



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Releases

No releases published

[Create a new release](#)

	laz380 and Chuck Witt	public release.	0aab4ce · yesterday	🕒 1 Commits
question-1	public release.	yesterday		
question-2	public release.	yesterday		
question-3	public release.	yesterday		
question-4	public release.	yesterday		
question-5	public release.	yesterday		
question-6	public release.	yesterday		
question-7	public release.	yesterday		
README.md	public release.	yesterday		



Gradescope Submission #20

Open

ryj579 opened this issue 2 days ago · 1 comment



ryj579 commented 2 days ago



Hello,

The submission instructions on the repo say to create a text file with the relevant information, but Gradescope only accepts either images or PDFs. Is it expected for us to create a word document or something with the name, hash, and link, or is there another way to submit to Gradescope I'm missing?

Thanks!



1



wcw398 commented 2 days ago • edited



Thanks - it looks like you are correct and the assignment type I selected in Gradescope is more restrictive than I realized. Gradescope also has a slightly different assignment type that accepts any file - I will use that for Homework 2 and beyond.

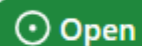
For Homework 1, please submit a PDF (e.g. use Word and print to PDF).

Apologies for the confusion.



1

SSH to the cluster #12



Open

chb1856 opened this issue 5 days ago · 4 comments



chb1856 commented 5 days ago



Hello. Setting up SSH from my machine to the cluster. I have the key downloaded, but cannot ping that host. Do I need to be on a VPN? And which hostname would be correct, so I am on a host that is safe to run jobs on?

Thanks!



wcw398 commented 5 days ago



Hello - I'd recommend holding off for now. The early experiments surfaced some issues we need to address with the cluster admins. We'll circulate some more detailed instructions if those issues can be resolved.



1

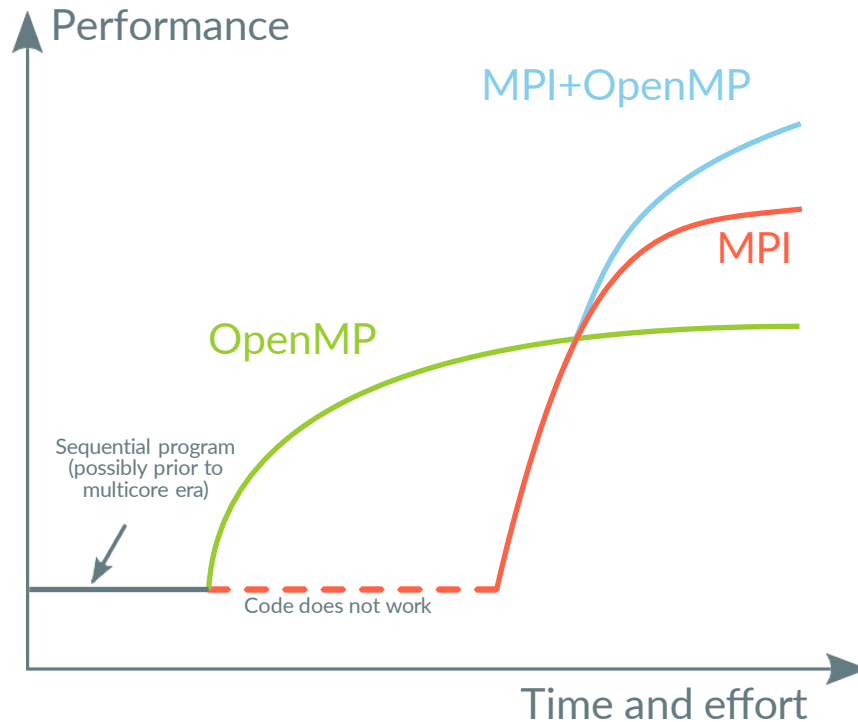
OpenMP: Open Multi-Processing



- Application Programming Interface (API) for explicit direct multi-threaded parallelism on shared memory architectures
 - Consists of 3 components:
 - Compiler directives
 - Runtime library calls
 - Environment variables
 - *A standard* widely used in industry and academic codes
-

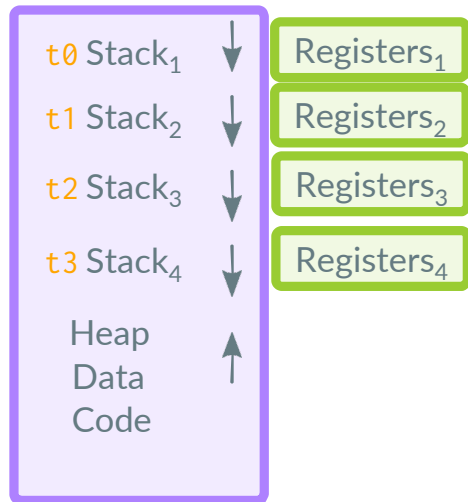


- *Is not* required to check your code for data races, deadlocks or whether your code conforms with the standard
- Is not designed for distributed memory systems
- Is not designed for parallel I/O



- OpenMP support can be added very easily to *existing* sequential code via compiler directives. (Not necessarily the case for alternatives)
- Compiler simply ignores these directives if it does not recognize them. To tell the compiler about these pragmas, you must add the `-fopenmp` flag at compile time (for GCC).
- Other programming models are also based on compiler directives, e.g., OpenACC.

RACE CONDITION IN COUNTER CLASS



Each thread maintains its own *state* of registers and program counter.

The `increment()` method in the `Counter` class results to memory.

```
void increment() { ++count_; }
```

When we disassemble this code and inspect the machine instructions that are executed for `++count_` we find essentially *3 instructions*:

```
mov edx, DWORD PTR [rax] ; load value at address rax into register edx (count_)
add edx, 1 ; add 1 to value in register edx, store result in edx
mov DWORD PTR [rax], edx ; write the value in edx to memory location rax
```

The data in registers `rax` and `edx` is *private to each thread*. If two threads execute this code, a **race condition** might look like this:

```
mov edx, DWORD PTR [rax] ; thread 0 loads current value of count_
mov edx, DWORD PTR [rax] ; thread 1 loads current value of count_
add edx, 1 ; thread 1 increments count_ and stores result in edx
add edx, 1 ; thread 0 increments count_ and stores result in edx
mov DWORD PTR [rax], edx ; thread 0 writes register edx to memory
mov DWORD PTR [rax], edx ; thread 1 writes register edx to memory (same value!)
```


EXAMPLE 1

OPENMP IN ACTION

$$\int_a^b \sin(x) dx \approx \left[\frac{\sin(a)}{2} + \frac{\sin(b)}{2} + \sum_{i=1}^{n-1} \sin(x_i) \right] \Delta x$$

EXAMPLE 2

NUMERIC INTEGRATION

Sift the Two's and Sift the Three's:
The Sieve of Eratosthenes.
When the multiples sublime,
The numbers that remain are Prime.

EXAMPLE 3

SIEVE OF ERATOSTHENES

NEXT TIME MPI

